

BANGLASOCIALBENCH: A Benchmark for Evaluating Socio-pragmatic and Cultural Alignment of LLMs in Bangladeshi Social Interaction

BANGLASOCIALBENCH: 用于评估大型语言模型在孟加拉国社交互动中的社会语言学和文化契合度的基准测试

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Abstract

摘要

Large Language Models have demonstrated strong multilingual fluency, yet fluency alone does not guarantee socially appropriate language use. In high-context languages, communicative competence requires sensitivity to social hierarchy, relational roles, and interactional norms that are encoded directly in everyday language. Bangla exemplifies this challenge through its three-tiered pronominal system, kinship-based addressing, and culturally embedded social customs. We introduce BANGLASOCIALBENCH, the first benchmark designed to evaluate sociopragmatic competence in Bangla through context-dependent language use rather than factual recall. The benchmark spans three domains: Bangla Address Terms, Kinship Reasoning, and Social Customs, comprising 1,719 culturally grounded instances written and verified by native Bangla speakers. We evaluate twelve contemporary LLMs in a zero-shot setting and observe systematic patterns of cultural misalignment. Models frequently default to overly formal address forms, fail to recognize multiple socially acceptable address pronouns, and conflate kinship terminology across religious contexts. Our findings show that sociopragmatic failures are often structured and non-random; for example, inappropriate addressing choices concentrate heavily in downward-hierarchy (Elder→Younger) and informal contexts. This reveals persistent limitations in how current LLMs infer and apply culturally appropriate language use in realistic Bangladeshi social interactions.

大型语言模型已经展现出强大的多语言流利度，但流利度本身并不能保证语言使用符合社会规范。在高语境语言中，交际能力要求对社会等级、关系角色以及互动规范具有敏感度，而这些规范直接编码在日常语言中。孟加拉语通过其三级代词系统、基于亲属关系的称呼方式以及文化嵌入的社会习俗，体现了这一挑战。我们引入了 BANGLASOCIALBENCH 基准，这是首个通过语境依赖的语言使用而非事实回忆来评估孟加拉语社会语言能力的基准。该基准涵盖三个领域：孟加拉语称呼用语、亲属关系推理和社会习俗，共包含 1,719 个由母语者撰写并验证的文化相关实例。我们在零样本设置下评估了十二种当代大型语言模型，并观察到系统性的文化错配模式。模型经常默认使用过于正式的称呼形式，未能识别多种社会上可接受的称呼代词，并且在宗教语境中混淆亲属术语。我们的研究表明，社会语言能力的失败往往是结构化的而非随机的；例如，不恰当的称呼选择在向下等级（长辈 $\rightarrow$ 晚辈）和非正式语境中尤为集中。这揭示了当前大型语言模型在现实孟加拉国社会互动中推断和应用文化适当语言使用方面的持续局限性。

1 Introduction

1 引言

Large Language Models (LLMs) have made rapid progress in multilingual generation, enabling fluent text production across a wide range of languages. However, fluency alone does not guarantee appropriate language use in social interaction, particularly in culturally grounded communi-

大型语言模型（LLMs）在多语言生成方面取得了快速进展，能够流畅地生成多种语言的文本。然而，流畅性本身并不能保证在社交互动中恰当使用语言，尤其是在文化基础的交流中，

প্রেক্ষাপট: একজন বৃদ্ধ যাত্রী মেট্রো স্টেশনে দাঁড়ানো কিশোর ছেলেকে বিনয়ের সাথে লাইনে দাঁড়াতে বলছেন।

Situation: An elderly passenger at a metro station is politely asking a teenage boy to stand in line.

In this context, how would the elderly person conventionally address the teenage boy **in Bangladeshi Culture**?

Answer in one word – apni/tumi/tui?

Setting Participants Ends Key Instruments Norms

Figure 1: [images/99ce9a3bdd1085807581cb81088798515cb1ab1ac0e194538d76f423aac13763.jpg](#)

[illegible]

在这一背景下，孟加拉国文化中，老人如何用传统的方式称呼年轻人？
用一句话回答——你/您/你？

In this context, how would the elderly person conventionally address the teenage boy in Bangladeshi Culture?

在此背景下，老年人通常如何用孟加拉国文化中的传统方式称呼这位青少年男孩？

Answer in one word — apni/tumi/tui?

用一个词回答 — apni/tumi/tui?

Figure 1: Prompt design grounded in Hymes’ SPEAKING model (Hymes, 1962). Each prompt operationalizes sociolinguistic context through explicit cues for setting, participants, gender, interactional goal, and social norms, allowing controlled evaluation of culturally appropriate Bangla Address Terms.

图 1: 基于 Hymes 的 SPEAKING 模型 (Hymes, 1962) 的提示设计。每个提示通过显式的线索 (如场景、

参与者、性别、互动目标和社会规范) 来操作化社会语言学背景, 从而允许对文化合适的孟加拉语称呼语进行受控评估。

cation. In many languages, meaning is shaped not only by what is said, but by how it is said, to whom, and in what context. When models fail to account for these interactional constraints, they may produce outputs that are linguistically correct yet socially inappropriate. This distinction aligns with the notion of communicative competence, which emphasizes contextual appropriateness alongside grammatical correctness (Hymes, 1964).

在许多语言中, 意义不仅由所说的内容塑造, 还由表达方式、对象以及所处的语境等因素共同决定。当模型未能考虑这些互动约束时, 可能会产生在语言上正确但社会上不恰当的输出。这一区别与“交际能力”的概念相一致, 该概念强调在语法正确的基础上, 还要考虑语境的适当性 (Hymes, 1964)。

This challenge is especially pronounced in high-context languages, where social relationships are encoded directly into everyday language use. Bangla, an Indo-Aryan language spoken by over 300 million people worldwide and used as the primary language by the vast majority of the population in Bangladesh (Sultana et al., 2023), exemplifies this phenomenon.¹ Rather than relying on

这一挑战在高语境语言中尤为明显, 因为在这些语言中, 社会关系直接编码在日常语言使用中。孟加拉语 (Bangla) 是一种印欧语系语言, 全球有超过 3 亿人使用, 是孟加拉国绝大多数人口的主要语言 (Sultana 等, 2023)。¹ 与依赖于

optional politeness markers, Bangla grammaticalizes social hierarchy through a three-tiered second-person pronominal system (apni, tumi, tui) and extensive kinship-based nominal addressing (Das, 1968a). Correct usage depends on factors such as power and formality, age difference, intimacy, emotion and inappropriate choices can sound rude or socially offensive despite correct propositional content (Uddin, 2019; Snigdha, 2022).

可选的礼貌标记, 孟加拉语通过一个三级第二人称代词系统 (apni, tumi, tui) 和广泛的基于亲属关系的名词称呼 (Das, 1968a) 来语法化社会等级。正确使用取决于诸如权力和正式程度、年龄差异、亲密程度、情感等因素, 即使命题内容正确, 不恰当的选择也可能听起来粗鲁或社会上冒犯 (Uddin, 2019; Snigdha, 2022)。

Despite the centrality of these phenomena to daily communication, existing evaluations only partially test whether models can reason over sociopragmatic distinctions in realistic interactional settings. Many existing multilingual evaluations rely on translated prompts, declarative questions, or factual recall tasks, which rarely require models to infer social roles or adapt language choices dynamically (Naous et al., 2024). As a result, models may appear culturally competent while still failing in direct social interaction.

尽管这些现象在日常交流中至关重要, 但现有的评估方法仅部分检验模型是否能够在现实的互动环境中进行社会语用区别的推理。许多现有的多语言评估依赖于翻译后的提示、陈述性问题或事实回忆任务, 这些任务很少要求模型推断社会角色或动态调整语言选择 (Naous 等, 2024)。因此, 模型可能在文化上显得胜任, 但在直接的社会互动中仍会失败。

To address this gap, we introduce BANGLASOCIALBENCH, a benchmark for evaluating sociopragmatic competence in Bangladeshi social contexts. The benchmark operationalizes language use as situated social behavior by embedding implicit sociopragmatic cues within realistic Bangla scenarios across three domains: Bangla Address Terms, Kinship Reasoning, and Social Customs. Using this framework, we evaluate twelve contemporary LLMs and observe consistent patterns of culturally inappropriate behavior. Across models, errors frequently arise from overuse of formal address forms, limited recognition of socially acceptable variation, and confusion in culturally grounded kinship usage, indicating systematic difficulties in resolving sociopragmatic ambiguity from context alone. In summary, our contributions are as follows:

为了解决这一差距, 我们引入了 BANGLASOCIALBENCH, 这是一个用于评估孟加拉国社会语境中社会语言能力的基准。该基准通过将隐含的社会语言线索嵌入到三个领域中的现实孟加拉语场景中, 将语言使

用视为具体的社会行为：孟加拉语称呼用语、亲属关系推理和社会习俗。利用这一框架，我们评估了十二个当代大型语言模型，并观察到一致的文化不适当行为模式。在这些模型中，错误经常源于正式称呼形式的过度使用、对社会可接受变体识别的限制以及基于文化背景的亲属关系使用中的混淆，这表明在仅凭上下文解决社会语言歧义方面存在系统性的困难。总之，我们的贡献如下：

- BANGLASOCIALBENCH: We introduce a benchmark that evaluates Bangla sociopragmatic competence through situated social behavior rather than static knowledge. It tests a model’s ability to infer hierarchy and intent across 1,719 human-written interactional scenarios. We make the benchmark publicly available to the research community.²
- We demonstrate that sociopragmatic failure in LLMs is often systematic rather
- BANGLASOCIALBENCH: 我们介绍了一个基准测试，通过情境中的社会行为而非静态知识来评估孟加拉语的社会语言能力。它测试了模型在 1,719 个由人类撰写的互动场景中推断层级和意图的能力。我们将这个基准测试公开给研究社区。²
- 我们证明了大语言模型在社会语言方面的失败往往是系统性的而非

than random and conditioned on variables like age and setting. We find that culturally inappropriate address pronoun choices concentrate in downward-hierarchy (Elder→Younger) and informal contexts, exposing structural weaknesses in reasoning about social power.

比随机选择更有条件地依赖于年龄和环境等变量。我们发现，文化上不恰当的称呼代词选择集中在向下等级（长辈 → 晚辈）和非正式语境中，暴露出在社会权力推理方面的结构性弱点。

- We uncover characteristic patterns of cultural misalignment in twelve contemporary LLMs, including pervasive over-politeness, probabilistic rigidity in address selection, and cross-religious kinship term conflation. These results suggest that models often rely on safe, default heuristics rather than socially grounded reasoning.

- 我们发现了十二个当代大型语言模型中文化错位的典型模式，包括普遍的过度礼貌、称呼选择中的概率刚性，以及跨宗教亲属称谓的混淆。这些结果表明，模型往往依赖于安全的默认启发式方法，而不是基于社会现实的推理。

2 Related Work

2 相关工作

2.1 Theoretical Frameworks and Factual Benchmarks

2.1 理论框架与事实基准

Recent work on cultural evaluation of LLMs distinguishes between culturally thin and culturally thick (Adilazuarda et al., 2024) conceptions of culture. Thin evaluations typically test recall of culture-related facts or alignment with aggregate statistics, offering broad but shallow signals of cultural awareness and often relying on outsider perspectives rather than lived experience. In contrast, culturally thick approaches view culture as an enacted and dynamic process, realized through language use in socially grounded interactional contexts (Zhou et al., 2025)

近期关于大型语言模型 (LLMs) 文化评估的研究区分了文化浅薄 (culturally thin) 和文化深厚 (culturally thick) 两种文化概念 (Adilazuarda 等, 2024)。浅薄的评估通常测试与文化相关的事实记忆或与汇总统计数据的一致性，提供了广泛但浅显的文化意识信号，往往依赖外部视角而非实际生活经验。相比之下，文化深厚的方法将文化视为一种实施和动态的过程，在社会基础的互动语境中通过语言使用得以实现 (Zhou 等, 2025)

A large class of benchmarks operationalizes culture as region-specific factual knowledge, testing models on history, geography, literature, and linguistic facts using curated multiple-choice questions. Examples include SaudiCulture (Ayash et al., 2025), SANSKRITI (Maji et al., 2025), CLiCK (Kim et al., 2024), and IndoCulture (Koto et al., 2024). In the Bangla context, BnMMLU (Joy, 2025) evaluates massive multitask understanding using academic materials, while BLUCK (Kabir et al., 2025) focuses on linguistic competence and cultural knowledge. While effective at measuring recall, they largely neglect the dynamic, interactional competence required for social acceptability, leaving a gap in evaluating how models navigate social deixis and pragmatic ambiguity.

一类大型基准测试将文化操作化为特定地区的事实性知识，通过精心设计的多项选择题，对模型在历史、地理、文学和语言事实等方面的能力进行测试。例如包括 SaudiCulture (Ayash 等, 2025)、SANSKRITI (Maji 等, 2025)、CLiCK (Kim 等, 2024) 和 IndoCulture (Koto 等, 2024)。在孟加拉语语境中，BnMMLU (Joy, 2025) 利用学术材料评估大规模多任务理解能力，而 BLUCK (Kabir 等, 2025) 则关注语言能力与文化知识。虽然这些测试在衡量回忆能力方面较为有效，但它们在很大程度上忽视了社交可接受性所需动态的、互动性的能力，从而在评估模型如何应对社会指示和语用模糊性方面留下了空白。

2.2 Value Alignment and Normative Reasoning

2.2 价值一致性与规范性推理

Another line of research evaluates cultural alignment through moral values and norms, comparing model responses against human population distributions derived from surveys or cultural indices (Zhao et al., 2024; AlKhamissi et al., 2024; Masoud et al., 2025). BengaliMoralBench (Ridoy et al., 2025) extends this approach to Bangla by auditing moral reasoning across culturally relevant subtopics. However, these approaches focus on abstract value alignment and largely overlook other semantic proxies of culture (Adilazuarda et al., 2024), particularly the realization of norms through interactional language use.

另一项研究通过道德价值观和规范来评估文化一致性，将模型的响应与来自调查或文化指数的人类人口分布进行比较 (Zhao 等, 2024; AlKhamissi 等, 2024; Masoud 等, 2025)。BengaliMoralBench (Ridoy 等, 2025) 将这种方法扩展到孟加拉语，通过审计与文化相关子主题中的道德推理来评估。然而，这些方法侧重于抽象的价值一致性，并在很大程度上忽略了文化其他语义代理 (Adilazuarda 等, 2024)，特别是通过互动语言使用实现规范。

2.3 Sociopragmatic and Linguistic Evaluation

2.3 社会语用与语言评估

Recent work has moved toward evaluating socio-pragmatic competence in language models, focusing on context-sensitive phenomena such as politeness, honorifics, and social etiquette. NormAd (Rao et al., 2025) evaluates whether models adapt outputs to sociocultural cues across scenarios, while PUB (Sravanthi et al., 2024) assesses pragmatic understanding across diverse contexts. The UNGGAH-UNGGUH benchmark (Farhansyah et al., 2025) evaluates register-sensitive honorific usage in Javanese, primarily through sentence-level transformations. In South Asian languages, prior work on Bangla and Hindi honorifics restricts analysis to third-person forms, explicitly excluding second-person pronouns due to their sociolinguistic complexity (Mukherjee et al., 2023).

近期的研究逐渐转向评估语言模型中的社会语言能力，重点关注诸如礼貌、敬语和社会礼仪等与语境相关的现象。NormAd (Rao 等, 2025) 评估模型是否能够根据不同的情境适应社会文化线索，而 PUB (Sravanthi 等, 2024) 则评估模型在不同语境下的语用理解能力。UNGGAH-UNGGUH 基准测试 (Farhansyah 等, 2025) 通过句子层面的转换，评估爪哇语中对敬语的敏感使用。在南亚语言中，针对孟加拉语和印地语敬语的先前研究限制分析仅限于第三人称形式，明确排除了第二人称代词，由于其社会语言学的复杂性 (Mukherjee 等, 2023)。

Several recent benchmarks examine sociopragmatic phenomena in high-context languages, where linguistic forms encode social hierarchy and interpersonal relations. For instance, KinshipQA (Sun and Kazakov, 2026) evaluates multi-hop reasoning over culturally grounded kinship structures derived from anthropological systems. TAAROFBENCH (Sadr et al., 2025) evaluates Persian taarof, a system of ritual politeness involving indirect offers and refusals.

近年来的一些基准测试研究了高语境语言中的社会语言现象，其中语言形式编码了社会等级和人际关系。例如，KinshipQA (Sun 和 Kazakov, 2026) 评估了基于人类学体系的、以文化为基础的亲属关系结构上的多跳推理。TAAROFBENCH (Sadr 等, 2025) 评估了波斯语的 taarof，这是一种涉及间接提议和拒绝的仪式性礼貌系统。

Related work also explores pragmatic phenomena in East Asian languages. The Mask of Civility (Zhang et al., 2026) evaluates Chinese mock politeness and highlights challenges in detecting mismatches between polite expressions and impolite contexts. Likewise, the KoBALT dataset (Shin et al., 2025) evaluates Korean across syntax, semantics, and morphology, revealing persistent difficulties with language-specific traits such as irregular verb conjugations and implicit interpersonal relations.

相关研究也探讨了东亚语言中的语用现象。《文明的面具》(Zhang 等, 2026) 评估了汉语中的委婉礼貌，并强调了在检测礼貌表达与不礼貌语境之间的不匹配时所面临的挑战。同样地，KoBALT 数据集 (Shin 等, 2025) 从句法、语义和形态学角度评估了韩语，揭示了诸如不规则动词变位和隐含的人际关系等语言特有特征所带来的持续困难。

2.4 Addressing the Gap

2.4 填补差距

Current multilingual evaluations often rely on English-centric or translated setups, which can obscure sociopragmatic distinctions encoded directly in native language use (Naous et al., 2024). As discussed, while recent efforts have introduced native Bangla benchmarks, they primarily target orthogonal capabilities. Datasets such as BnMMLU (Joy, 2025) and BLUCK (Kabir et al., 2025) focus heavily on massive multitask factual recall and static linguistic knowledge. Similarly, BengaliMoralBench (Ridoy et al., 2025) audits abstract moral reasoning, and prior work on Bangla honorifics restricts analysis strictly to third-person forms (Mukherjee et al., 2023).

当前的多语言评估通常依赖于以英语为中心或翻译设置，这可能会掩盖直接体现在原语言使用中的社会语用区别 (Naous 等, 2024)。正如所讨论的，尽管最近的努力引入了本土孟加拉语基准，但它们主要针对正交能力。例如，BnMMLU (Joy, 2025) 和 BLUCK (Kabir 等, 2025) 等数据集高度关注大规模多任务事实回忆和静态语言知识。同样，BengaliMoralBench (Ridoy 等, 2025) 审核抽象道德推理，而关于孟加拉语敬语的先前研究严格限制分析仅限于第三人称形式 (Mukherjee 等, 2023)。

While effective in their domains, existing benchmarks neglect the dynamic interactional competence required for everyday communication. In contrast, communicative competence in Bangla relies heavily on sensitivity to social hierarchy, relational roles, and interactional norms, requiring models to jointly reason over hierarchical address forms, complex kinship structures, and culturally embedded social customs. To address this, BANGLASOCIALBENCH introduces a culturally thick framework evaluating sociopragmatic competence in realistic Bangladeshi settings.

尽管现有基准在各自领域内是有效的，但它们忽视了日常交流中所需的动态互动能力。相比之下，孟加拉语的交际能力高度依赖于对社会等级、关系角色和互动规范的敏感性，要求模型能够共同推理层级称呼形式、复杂的亲属结构以及文化嵌入的社会习俗。为了解决这一问题，BANGLASOCIALBENCH 引入了一个文化丰富的框架，用于在现实的孟加拉国环境中评估社会语言能力。

3 Cultural Context and Overview of BANGLASOCIALBENCH

3 文化背景与 BANGLASOCIALBENCH 概览

BANGLASOCIALBENCH evaluates LLM sociopragmatic competence across three domains: Address Terms, Kinship Reasoning, and Social Customs.

BANGLASOCIALBENCH 评估大型语言模型（LLM）在三个领域中的社会语言能力：称呼用语、亲属关系推理和社会习俗。

3.1 Bangla Address Terms

3.1 孟加拉语地址用语

Address forms are a primary mechanism through which Bangladeshi speakers manage social relations in interaction. Bangla employs a three-way second-person pronominal system that encodes different levels of social distance and respect. The form *apni* expresses the highest level of formality and is typically used for elders, authority figures, or strangers. The form *tumi* conveys neutral familiarity among peers or polite downward rela-

地址形式是孟加拉语使用者在互动中管理社会关系的主要机制。孟加拉语采用三重第二人称代词系统，编码了不同层次的社会距离和尊重。形式“*apni*”表达了最高层次的正式程度，通常用于长辈、权威人物或陌生人。形式“*tumi*”则在同龄人之间传达中性熟悉感，或用于礼貌的下级关系中。

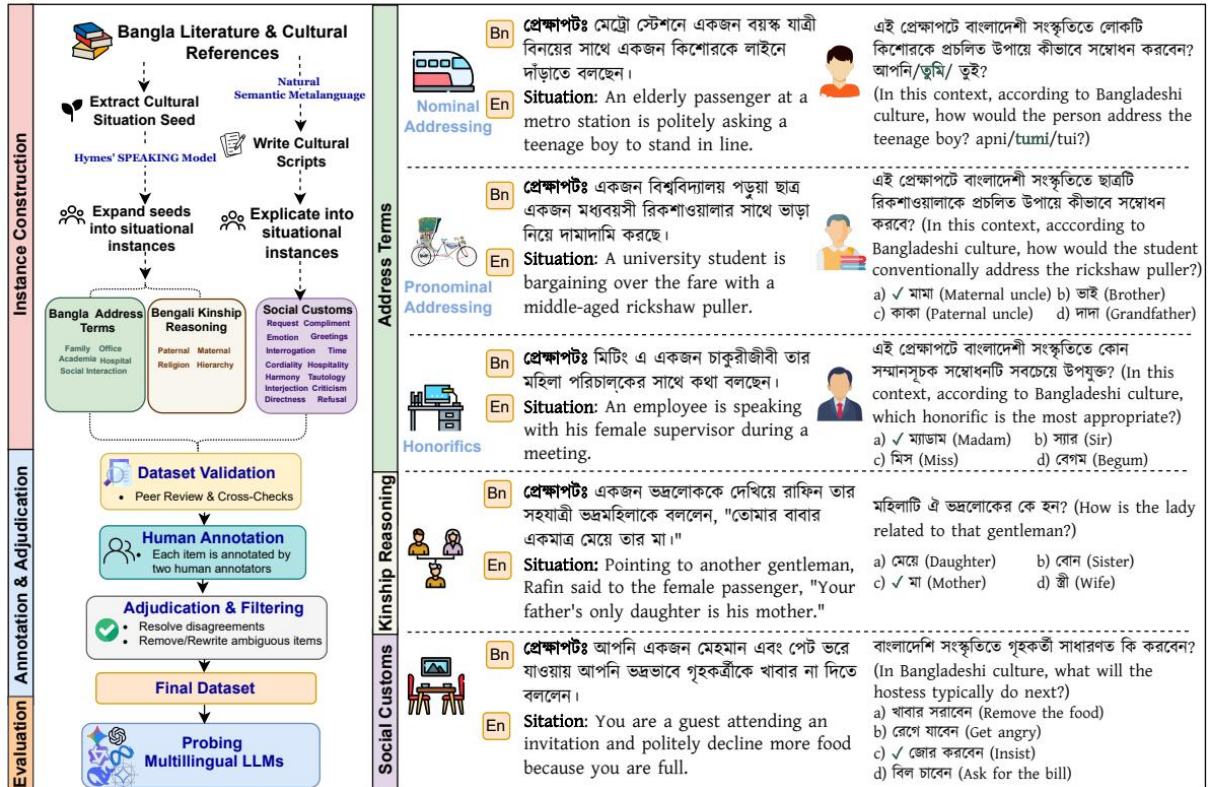


Figure 2: images/c0759db7ba37bf189876f401de8de11360bb7104b872eb118e42926ba89a6ff9.jpg

Figure 2: Dataset creation pipeline for BANGLASOCIALBENCH. The English prompts displayed in the diagram are translated for illustrative purposes; all model evaluations were conducted

exclusively using Bangla prompts

图 2: BANGLASOCIALBENCH 数据集创建流程。图中显示的英文提示语为便于说明而进行了翻译；所有模型评估均仅使用孟加拉语提示语进行

tions. The form *tui* signals intimacy and is commonly used among close friends, siblings, or when addressing children. Because these forms directly encode hierarchy and relational distance, incorrect selection can sound rude or socially inappropriate despite grammatically correct content.

这些形式体现了亲密关系，通常用于亲密的朋友、兄弟姐妹之间，或在称呼孩子时使用。由于这些形式直接编码了等级和关系的距离，即使内容在语法上是正确的，选择不当也可能听起来粗鲁或社会上不恰当。

Beyond pronouns, Bangla widely uses kinship-based nominal addressing, where kinship terms are used both for relatives and for socially proximate non-relatives to signal age, gender, and hierarchy (Sultana et al., 2023). Honorific titles further refine these distinctions across professional, religious, and everyday settings (Das, 1968b). Together, these resources require speakers to infer appropriate address forms from contextual cues such as age difference, social role, and interactional goal.

除了代词之外，孟加拉语广泛使用基于亲属关系的名词称呼方式，其中亲属称谓既用于亲属，也用于社会关系密切但非亲属的人，以表明年龄、性别和等级（Sultana 等，2023）。敬语头衔进一步在专业、宗教和日常场合中细化这些区别（Das, 1968b）。这些资源共同要求说话者从年龄差异、社会角色和互动目标等语境线索中推断出适当的称呼形式。

To operationalize these distinctions, prompts are constructed following Hymes’ Ethnography of Communication and the SPEAKING framework (Hymes, 1964). Each instance specifies participants, setting, gender and interactional intent so that the correct address form must be inferred from sociopragmatic context rather than explicit lexical hints (Figure 1). Additional cultural context is pro-

为了具体化这些区别，提示语是根据 Hymes 的交流民族志和 SPEAKING 框架（Hymes, 1964）构建的。每个实例都指定了参与者、场景、性别和交流意图，从而使正确的称呼形式必须从社会语用语境中推断出来，而不是从显性的词汇提示中得出（图 1）。此外，还提供了额外的文化背景信息。

vided in Appendix A, and example instances appear in Tables 4-9.

见附录 A，示例实例见表 4-9。

3.2 Bengali Kinship Reasoning

3.2 孟加拉语亲属关系推理

Bangla kinship terminology encodes detailed distinctions that require structured relational reasoning. Unlike English, which often collapses multiple relations into broad categories such as “uncle”, Bangla differentiates kinship roles based on lineage (paternal vs. maternal) and generational order (Kolenda and Aziz, 1981; Das, 1968b). For example, the language distinguishes *chacha* (father’s brother) from *mama* (mother’s brother), among many other lineage-specific terms.

孟加拉语的亲属称谓编码了需要结构化关系推理的详细区别。与英语不同，英语常常将多个关系归入如“叔叔”这样的宽泛类别中，而孟加拉语则根据血缘关系（父系与母系）和世代顺序（Kolenda 和 Aziz, 1981; Das, 1968b）来区分亲属角色。例如，该语言区分了“*chacha*”（父亲的兄弟）和“*mama*”（母亲的兄弟），还有许多其他特定于血缘关系的术语。

To evaluate whether models can infer such relations, we construct a dataset of 345 kinship reasoning puzzles based on blood relations. Each instance presents a multi-hop lineage description and requires the model to identify the correct kinship term. The dataset spans maternal, paternal, and affinal relations, with distractor options designed to test sensitivity to lineage direction

and generational hierarchy.

为了评估模型是否能够推断出此类关系，我们构建了一个包含 345 个基于血缘关系的亲属推理谜题的数据集。每个实例都提供一个多跳的家族谱描述，要求模型识别正确的亲属关系术语。该数据集涵盖了母系、父系和姻亲关系，干扰选项的设计旨在测试模型对家族谱方向和世代层级的敏感度。

3.3 Bengali Social Customs

3.3 孟加拉国的社会习俗

The Social Customs domain evaluates culturally grounded expectations that govern appropriate responses and behavior in everyday Bangladeshi interaction. Each instance presents a realistic scenario followed by four candidate responses that are grammatically plausible but differ in pragmatic appropriateness. Models must select the response that best aligns with commonly shared social norms.

社会习俗领域评估了基于文化背景的期望，这些期望规范了日常孟加拉国互动中适当的行为和反应。每个实例都呈现一个现实场景，随后是四个语法上合理但语用适当性不同的候选回答。模型必须选择最符合普遍共享社会规范的回答。

Scenarios are derived from ethnographic descriptions of Bangladeshi social interaction and iteratively validated by native Bangla annotators. To structure these situations, we use cultural scripts (Goddard and Wierzbicka, 2004), formalized using Natural Semantic Metalanguage (Wierzbicka, 1996). This process identifies fourteen recurring scripts that capture interactional expectations (shown in Table 1).

场景来源于对孟加拉国社会互动的民族志描述，并通过本地孟加拉语标注者反复验证。为了结构化这些情境，我们使用文化脚本 (Goddard 和 Wierzbicka, 2004)，并使用自然语义元语言 (Wierzbicka, 1996) 进行形式化。这一过程识别出十四种反复出现的脚本，这些脚本捕捉了互动期望（如表 1 所示）。

3.4 BANGLASOCIALBENCH Development

3.4 BANGLASOCIALBENCH 开发

3.4.1 Instance Writing

3.4.1 实例写作

Instance creation began with a cultural situation brainstorming phase conducted by six native Bangla annotators who are lifelong residents of Bangladesh and have extensive exposure to everyday Bangladeshi social interaction. Annotators first generated short descriptions of socially plausible interactions drawn from daily life.

实例创建始于由六名本地孟加拉语标注员进行的文化情境头脑风暴阶段。这些标注员是长期居住在孟加拉国的孟加拉语母语者，并且对日常孟加拉国社会互动有广泛的了解。标注员首先根据日常生活生成了简短的、社会上合理的互动描述。

To ensure cultural grounding, annotators were allowed to consult Bangladeshi literary works (short stories, drama, or novels) and ethnographic or sociological sources for inspiration. They were explicitly instructed not to copy text, characters, or narrative structures. All instances were written independently from scratch to reduce the possibility of memorization or data leakage from LLM training corpora.

为确保文化根基，注释者被允许参考孟加拉国的文学作品（短篇小说、戏剧或小说）以及民族志或社会学

资料以获取灵感。他们被明确指示不得复制文本、人物或叙事结构。所有实例均独立从头撰写，以减少记忆或从大型语言模型训练语料库中泄露数据的可能性。

These initial situation seeds were then expanded in a domain-specific manner. For Address Terms, scenarios were systematically varied along sociopragmatic variables such as participant age, gender, key, end, and setting (Hymes, 1964). For Kinship Reasoning, annotators constructed relational puzzles requiring inference over Bangladeshi kinship structures, using external materials only as conceptual reference. For Social Customs, recurring cultural practices were abstracted into shared cultural scripts and instantiated as multiple-choice scenarios reflecting conventional Bangladeshi expectations. Detailed annotator recruitment and instance creation procedures are described in Appendix C.

这些初始的情境种子随后以领域特定的方式进行了扩展。对于地址用语，场景沿社会语用变量（如参与者年龄、性别、关键、结束和情境）系统性地变化（Hymes, 1964）。对于亲属关系推理，标注者构建了需要对孟加拉国亲属结构进行推理的关系谜题，仅使用外部材料作为概念参考。对于社会习俗，反复出现的文化实践被抽象为共享的文化脚本，并实例化为多个选择场景，反映了传统的孟加拉国期望。详细的标注者招募和实例创建流程在附录 C 中进行了描述。

Category	Subcategory	#
Bangla Address Term: Pronominal	Family	140
Office	100	
Hospital	50	
Academia	100	
Social Interaction	200	
Bangla Address Term: Nominal	Kinship	152
Social Addressing	155	
Honorifics	85	
Bengali Kinship Reasoning	Mixed Relations	345
Bengali Social Customs	Request	20
Interrogation	14	
Compliment	40	
Refusal	20	
Tautology	20	
Interjection	20	
Indirect Directness	20	
Hospitality	40	
Greetings	20	
Emotion	20	
Harmony	20	
Cordiality	78	
Criticism	40	
Time	20	
Total		1719

类别	子类别	#
孟加拉地址用语：代词	家庭	140

Office	100	
医院	50	
学术界	100	
社交互动	200	
孟加拉地址术语：名词	亲属关系	152
社交地址	155	
尊称	85	
孟加拉语亲属关系推理	复杂关系	345
孟加拉社会习俗	请求	20
审问	14	
赞美	40	
拒绝	20	
同义反复	20	
感叹词	20	
间接的直接性	20	
hospitality	40	
问候	20	
情绪	20	
和谐	20	
礼貌	78	
批评	40	
时间	20	
总计		1719

Table 1: Distribution of instances in BANGLASOCIALBENCH across address terms, kinship reasoning, and social customs domains.

表 1: BANGLASOCIALBENCH 中实例在地址称谓、亲属关系推理和社会习俗领域的分布情况。

3.4.2 Dataset Quality Control

3.4.2 数据集质量控制

Annotation Procedure Each instance was independently annotated by two annotators. Annotation guidelines emphasized selecting responses that reflect widely shared Bangladeshi social expectations rather than individual preferences. For pronominal address-term instances, annotators were allowed to specify both a primary and, when pragmatically acceptable, a secondary form.

注释过程每个实例均由两名注释者独立进行注释。注释指南强调选择反映孟加拉国普遍社会期望的回应，而不是个人偏好。对于代词称呼用语的实例，注释者可以指定一个主要形式，并在语用上可接受的情况下指定一个次要形式。

Inter-Annotator Agreement Measurement Agreement was measured using Cohens κ (Cohen, 1960). For single-answer subsets, the selected label was used directly. For pronominal subsets allowing two acceptable forms, agreement was computed using the primary label.

Inter-Annotator Agreement Measurement 同意度是使用 Cohens κ (Cohen, 1960) 进行测量的。对于单选子集，直接使用所选标签。对于允许两种可接受形式的代词子集，使用主要标签计算同意度。

Across all applicable subsets, the global standard κ was 0.82, indicating strong agreement. For tasks allowing two valid pronouns, we additionally computed a binary κ (Artstein and Poesio, 2008) by treating each pronoun as an independent binary

在所有适用的子集上，全球标准 κ 为 0.82，表明有很强的一致性。对于允许使用两个有效代词的任务，我们还通过将每个代词视为独立的二元变量，额外计算了一个二元 κ (Artstein 和 Poesio, 2008)。

decision and macro-averaging the results. This produced a score of 0.77. Detailed annotation protocols and subcategory-wise statistics are reported in Table 10 of Appendix D.

决策并进行宏平均处理结果。这产生了 0.77 的得分。详细的注释协议和子类别的统计数据在附录 D 的表 10 中报告。

Adjudication and Label Consolidation All annotation disagreements were resolved through a two-stage process consisting of peer discussion followed by verification by senior annotators. Instances judged to be ambiguous, underspecified, or culturally contentious were revised or removed. For the Kinship Reasoning subset, which involves deterministic reasoning, standard agreement metrics were not applicable. Because each item corresponds to a single ground truth, any disagreement was treated as an annotation error and resolved through third-party adjudication.

裁决和标签整合所有注释分歧都通过一个两阶段的流程解决，包括同行讨论后由高级注释员进行验证。被判定为模棱两可、描述不明确或文化上存在争议的实例会被修订或删除。对于涉及确定性推理的 Kinship Reasoning 子集，标准的协议度量指标并不适用。因为每个项目对应一个单一的真实标签，任何分歧都会被视为注释错误，并通过第三方裁决来解决。

Dataset Filtering Approximately 13% of initially drafted instances were excluded during quality control due to unresolved ambiguity or insufficient cultural grounding.

数据集筛选大约 13% 的最初草拟实例在质量控制过程中因未解决的歧义或文化基础不足而被排除。

4 Experiments

4 实验

Models We evaluate a total of 12 LLMs, comprising six closed-source and six open-weight models. The closed-source models include GPT-4o-mini, GPT-4o, Gemini Flash 2.0, Gemini Flash 2.5, Claude Haiku 3.5, Claude Sonnet 4. The open-weight models include LLaMA 3 8B Instruct, LLaMA 3.3 70B Instruct, Gemma 3 12B, Gemma 3 27B, Qwen 2.5 72B Instruct and DeepSeek V3.1. Together, these models span diverse scales, paradigms, and levels of Bangla support.

我们评估了总共 12 个 LLMs，包括六个闭源模型和六个开源模型。闭源模型包括 GPT-4o-mini、GPT-4o、Gemini Flash 2.0、Gemini Flash 2.5、Claude Haiku 3.5、Claude Sonnet 4。开源模型包括 LLaMA 3 8B Instruct、LLaMA 3.3 70B Instruct、Gemma 3 12B、Gemma 3 27B、Qwen 2.5 72B Instruct 和 DeepSeek V3.1。这些模型涵盖了多样化的规模、范式以及孟加拉语支持水平。

Evaluation Protocol All models are evaluated in a zero-shot setting using prompts written entirely in Bangla, without task-specific fine-tuning or in-context examples. A standardized prompt template is used across all models for comparability. Each prompt presents a contextual scenario and asks the model to select the culturally appropriate response in Bangladeshi social context. Decoding is performed with temperature set to 0 to ensure deterministic outputs. All task instructions are provided within the user prompt. Example prompts are shown in Appendix D.5.1.

评估协议所有模型均在零次设置下进行评估，使用完全用孟加拉语编写的提示，而不进行任务特定的微调

或上下文示例。为了可比性，所有模型均使用标准化的提示模板。每个提示都提供一个情境场景，并要求模型在孟加拉国社会背景下选择文化上合适的回应。解码时将温度设置为 0 以确保确定性输出。所有任务指令均包含在用户提示中。示例提示见附录 D.5.1。

Dual Acceptable Forms For Pronominal Addressing within (1) Bangla Address Terms domain, each instance is annotated with a primary pronoun and, when applicable, a secondary acceptable pronoun. This reflects the sociopragmatic reality that multiple address forms may be culturally appropriate for the same interactional context. Accordingly, a model response is counted as correct if it matches either the primary or the secondary acceptable pronoun, ensuring that culturally appropriate variation in address forms is not penalized.

双可接受的代词称呼形式在（1）孟加拉语称呼用语领域中，每个实例都标注有主要代词，并在适用时标注一个次要可接受的代词。这反映了社会语言学现实，即在相同的交际情境中，可能存在多种文化上合适的称呼形式。因此，如果模型的回答与主要代词或次要可接受的代词之一相符，则视为正确，从而确保在称呼形式上的文化适当性变化不会受到惩罚。

Because some instances specify two acceptable pronouns, we report an instance weighted random baseline of 42.2% for this subset of the benchmark. (Details can be found in Appendix D.5.2)

由于一些实例指定了两个可接受的代词，我们针对该基准测试的子集报告了一个实例加权随机基线，准确率为 42.2%。（详情请参见附录 D.5.2）

Single Best Answer For (2) Kinship Reasoning, Nominal Addressing within (1) Bangla Address Forms, and (3) Social Customs, evaluation is conducted using exact match accuracy over four-option Multiple-Choice Questions. A model response is considered correct if it selects the culturally appropriate option specified in the annotated reference. Under this setting, a uniform random baseline achieves 25% accuracy.

针对（2）亲属关系推理、（1）孟加拉语称呼形式中的名义称呼，以及（3）社会习俗，评估是通过四选项多项选择题的精确匹配准确率来进行的。如果模型响应选择了注释参考中指定的文化上合适的选项，则认为该响应是正确的。在此设置下，一个统一的随机基线准确率为 25%。

4.1 Evaluation of Language Models on BANGLASOCIALBENCH

4.1 在 BANGLASOCIALBENCH 上评估语言模型

Performance varies across domains and model size As shown in Table 2 and Figure 3, model performance varies substantially across domains, with Kinship Reasoning posing the greatest challenge. Accuracy ranges from 31.88% (LLaMA 3 8B) to 92.46% (Gemini 2.5 Flash) on Kinship Reasoning, compared to 46.23-81.16% for Address Terms and 46.67-89.49% for Social Customs. Results also reveal a consistent model-scale effect within model families. For example, within the LLaMA family, accuracy increases from 46.23% (8B) to 63.24% (70B) on Address Terms and from 31.88% to 51.01% on Kinship Reasoning. A similar trend appears for Gemma models (54.99% to 73.42% on Address Terms). To ensure that subcategory imbalance (e.g., 200 instances for Social Interaction vs. 50 for Hospital) does not skew our findings, we verified that micro-averaged and macro-averaged accuracies across the Address Terms domain deviate by < 2%, confirming our overall metrics accurately reflect true model competence.

性能在不同领域和模型规模之间存在差异。如表 2 和图 3 所示，模型在不同领域的性能差异显著，其中亲属关系推理（Kinship Reasoning）最具挑战性。在亲属关系推理任务中，准确率从 LLaMA 3 8B 的 31.88% 到 Gemini 2.5 Flash 的 92.46% 不等，相比之下，地址术语（Address Terms）的准确率范围为 46.23%-81.16%，社会习俗（Social Customs）的准确率范围为 46.67%-89.49%。结果还揭示了在同一模型家族中存在一致的模型规模效应。例如，在 LLaMA 家族中，地址术语的准确率从 8B 的 46.23% 提升到 70B 的 63.24%，而在亲属关系推理任务中，准确率从 31.88% 提升到 51.01%。Gemma 模型也呈现出类似的趋势（地址术语的准确率从 54.99% 提升到 73.42%）。为确保子类别的不平衡（例如，社交互动有 200 个实例，而医院仅有 50 个实例）不会影响我们的研究结果，我们验证了地址术语领域中微平均和宏平均准确率之间的差异小于 2%，从而确认我们的总体指标准确反映了模型的真实能力。

Overall, larger proprietary models perform strongest, with Gemini 2.5 Flash achieving the highest accuracy on Address Terms and Kinship Reasoning, while Claude Sonnet 4 performs best on Social Customs. These results suggest that sociopragmatic competence improves with scale but remains uneven across domains.

总体而言，较大的专有模型表现最为出色，其中 Gemini 2.5 Flash 在地址术语和亲属关系推理方面取得了最高的准确性，而 Claude Sonnet 4 在社会习俗方面表现最佳。这些结果表明，社会语言能力随着模型规模的增加而提高，但在不同领域之间仍存在不均衡。

Model	Address Term	Kinship Reasoning	Social Customs
GPT-4o mini	65.99	49.86	77.43
GPT-4o	78.51	67.83	86.15
Gemini 2.0 Flash	76.17	68.12	85.13
Gemini 2.5 Flash	81.16	92.46	86.41
Claude Haiku 3.5	61.91	44.06	71.02
Claude Sonnet 4	75.46	88.98	86.67
LLaMA 3 8B Instruct	46.23	31.88	42.82
LLaMA 3.3 70B Instruct	63.24	51.01	81.54
Gemma 3 12B	54.99	42.61	76.15
Gemma 3 27B	73.42	48.12	81.02
Qwen 2.5 72B Instruct	64.77	48.70	77.18
DeepSeek V3.1	64.77	64.06	83.33

模型	称呼用语	亲属关系推理	社会习俗
GPT-4o mini	65.99	49.86	77.43
GPT-4o	78.51	67.83	86.15
Gemini 2.0 闪充	76.17	68.12	85.13
Gemini 2.5 Flash	81.16	92.46	86.41
Claude Haiku 3.5	61.91	44.06	71.02
莎士比亚十四行诗第四首	75.46	88.98	86.67
LLaMA 3 8B Instruct	46.23	31.88	42.82
LLaMA 3.3 70B Instruct	63.24	51.01	81.54
Gemma 3 12B	54.99	42.61	76.15
Gemma 3 27B	73.42	48.12	81.02
Qwen 2.5 72B Instruct	64.77	48.70	77.18
DeepSeek V3.1	64.77	64.06	83.33

Table 2: Overall zero-shot accuracy (%) across domains in BANGLASOCIALBENCH.

表 2: BANGLASOCIALBENCH 中跨领域的总体零样本准确率 (%)。

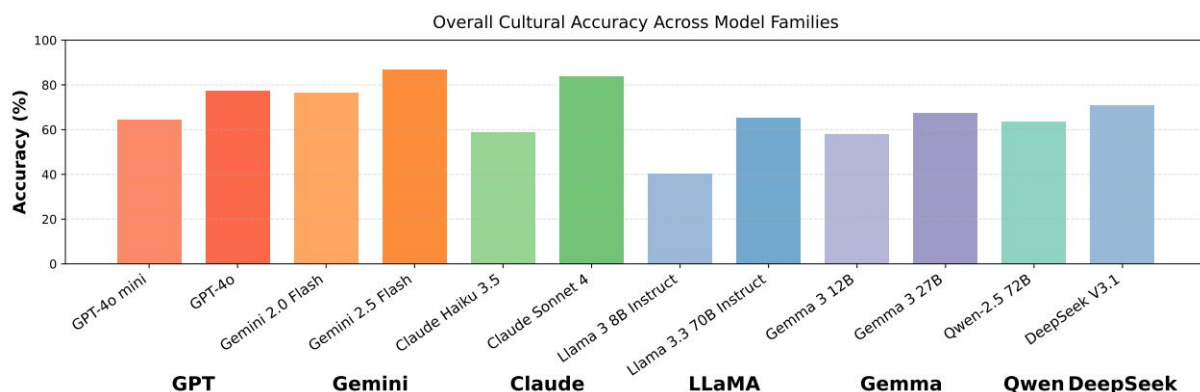


Figure 3: images/ec2ccac92cf352f3307dc8d5ad6ba34194c5d4117342221429a52fc1e523b538.jpg

Figure 3: Overall benchmark accuracy of evaluated LLMs on BANGLASOCIALBENCH, computed as the macro-average of performance across Address Terms, Kinship Reasoning, and Social Customs.

图 3：在 BANGLASOCIALBENCH 上评估的 LLMs 的整体基准准确率，计算方式为 Address Terms、Kinship Reasoning 和 Social Customs 各项性能的宏平均值。

Most models exhibit systematic over-politeness As shown in Figure 4, most models display a clear asymmetry in culturally inappropriate pronominal choices. A binomial test comparing over- and under-politeness errors shows a statistically significant imbalance for most models ($p < 0.05$), with over-politeness occurring substantially more frequently. Several models, including DeepSeek V3.1 and Gemma 3 12B, frequently default to the highly formal pronoun *apni*, even in contexts where *tumi* or *tui* are culturally preferred. In contrast, under-politeness errors are rare across all model families. This pattern suggests that models adopt a conservative strategy, favoring safer or more polite forms rather than adapting flexibly to contextual cues such as age difference, intimacy, or interactional setting. This may reflect training data skew, since high-resource Bangla sources predominantly use formal registers.

大多数模型表现出系统性的过度礼貌。如图 4 所示，大多数模型在文化不合适的代词选择上表现出明显的不对称性。一项比较过度礼貌和欠礼貌错误的二项式检验显示，大多数模型存在统计学上显著的不平衡 ($p < 0.05$)，其中过度礼貌的出现频率明显更高。包括 DeepSeek V3.1 和 Gemma 3 12B 在内的多个模型，即使在应使用 *tumi* 或 *tui* 的文化偏好情境中，也经常默认使用高度正式的代词 *apni*。相比之下，所有模型家族中欠礼貌的错误都较为罕见。这种模式表明，模型采用了一种保守的策略，更倾向于选择更安全或更礼貌的形式，而不是灵活地根据年龄差异、亲密程度或互动场景等上下文线索进行调整。这可能反映了训练数据的偏差，因为高资源的孟加拉语来源主要使用正式语体。

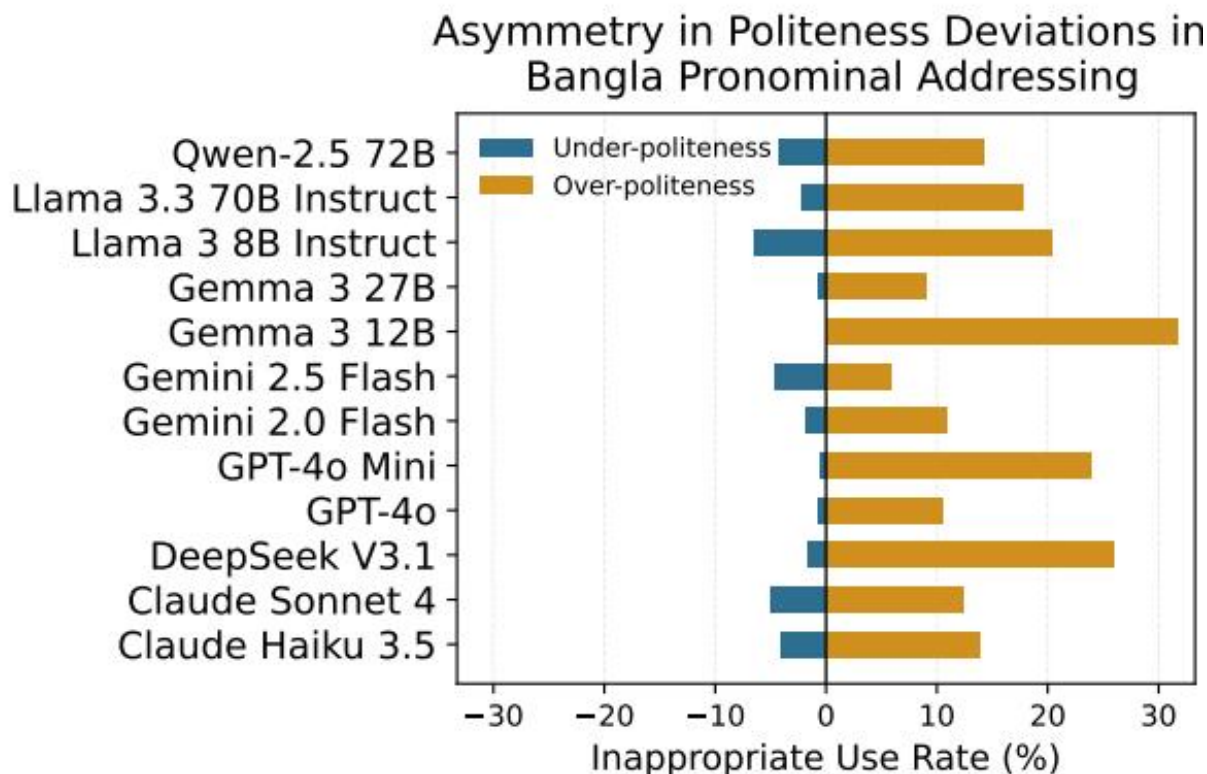


Figure 4: images/d5854626e8312766b582e0297da017b4b4bf8a8af4c7cd4441e039f42c4622b0.jpg

Figure 4: Asymmetry in inappropriate politeness use across LLMs in Bangla pronominal addressing, with over-politeness occurring substantially more frequently than under-politeness.

图 4：在孟加拉语代词称呼中，不同大语言模型在不恰当礼貌使用方面的不对称性，过度礼貌的使用频率明显高于不足礼貌的使用。

4.2 Analyzing Contextual Effects in Culturally Inappropriate Predictions

4.2 分析文化不适应预测中的上下文影响

Sociopragmatic factors linked to inappropriate Address Pronoun choice To identify sys-

与不恰当的称呼代词选择相关的社会语言学因素为识别 sys-

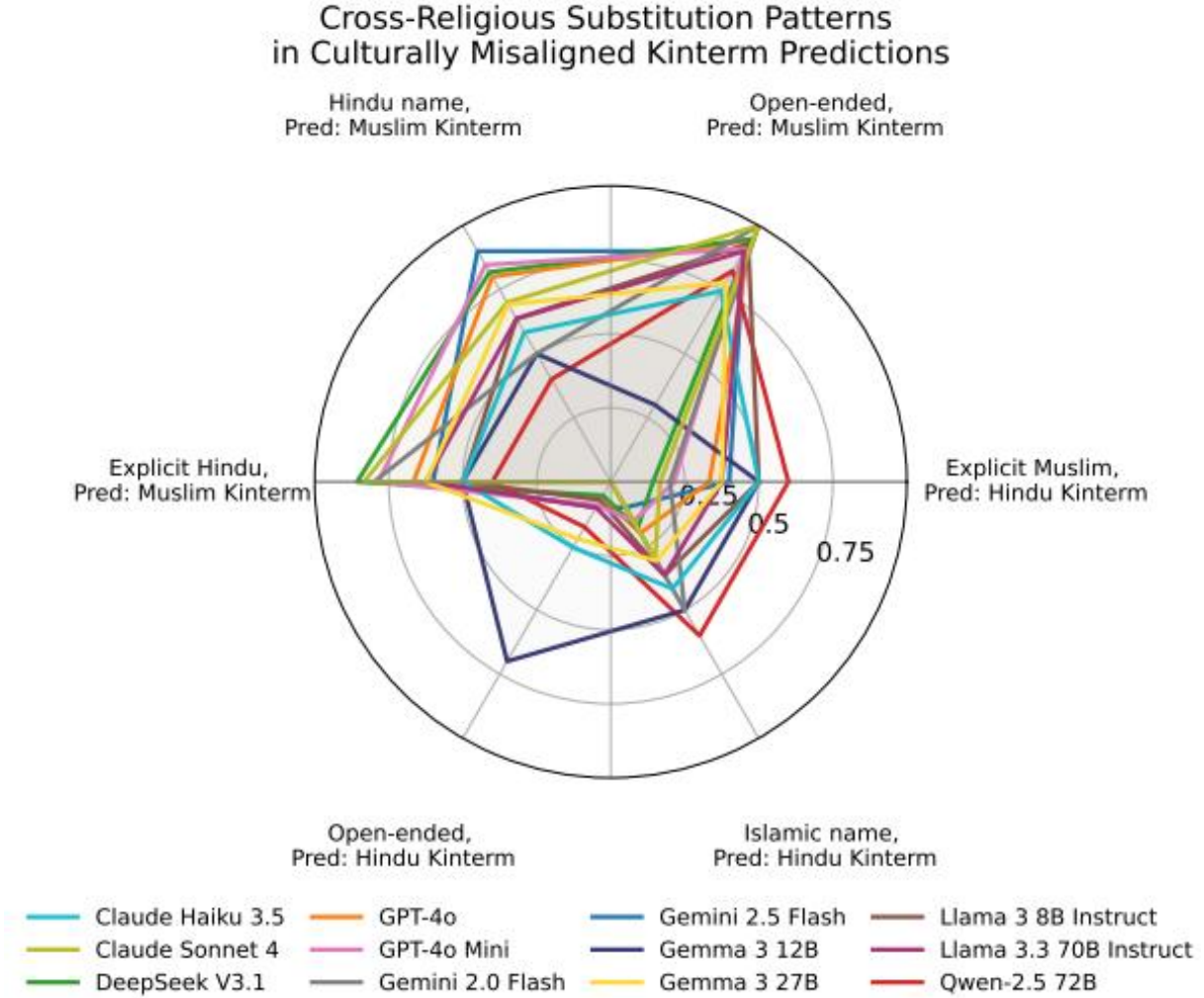


Figure 5: images/2d78b50e8c714c8c4c6d3233028164cfa7f9847c0234a4c7cb83b864ec77dc6f.jpg

Figure 5: Directional cross-religious kinship term misalignment. Proportions of culturally inappropriate kinship term substitutions across explicit identity cues, implicit cues, and open-ended prompting. Misalignment is more pronounced toward substituting Muslim-associated kinterms in Hindu-marked contexts.

图 5：方向性跨宗教亲属称谓错配。在显性身份线索、隐性线索和开放式提示中，文化不合适的亲属称谓替换的比例。错配在将与穆斯林相关的亲属称谓替换为印度教标记情境时更为明显。

tematic sources of cultural misalignment, we conduct Pearson’s χ^2 tests of independence between culturally inappropriate pronominal address term choices and five sociopragmatic factors: Participant Age, Setting, Key, End, and Participant Gender, evaluated over $N = 590$ instances per model. Standardized residual analysis further indicates that culturally inappropriate address choices concentrate in downward-hierarchy age relations (Elder $\rightarrow$ Younger) and informal or unstructured settings, whereas upward-hierarchy interactions and formal contexts exhibit relative robustness. Full statistics and dataset balance details are reported in Appendix D.5.3 and Appendix B (Figures 8, 9).

针对文化不匹配的系统性来源，我们对文化不合适的代词称呼用语选择与五个语用社会因素（参与者年龄、场合、关键点、结束和参与者性别）之间的独立性进行了皮尔逊的 χ^2 检验，每种模型评估了 $N = 590$ 个实例。标准化残差分析进一步表明，文化不合适的称呼选择主要集中在向下等级关系（长辈 $\rightarrow$ 晚辈）和非正式或非结构化场合中，而向上等级互动和正式场合则表现出相对的稳健性。完整的统计信息和数据集平

衡详情在附录 D.5.3 和附录 B（图 8、9）中报告。

Language Models conflate kinship terms across religious identities Culturally inappropriate predictions display clear directional asymmetries across models. In all experiments, models were instructed to answer according to Bangladeshi cultural practices across multiple prompt strategies, including explicit religious identity markers (e.g., “a Muslim person”), religiously indicative personal names (e.g., “Arnab Roy,” a culturally Hindu name), and open-ended prompts without overt religious cues. Several models, including GPT-4o, DeepSeek V3.1, and Gemini 2.5 Flash, show a stronger tendency to substitute Muslim-associated kinship terms in explicitly or implicitly Hindu contexts, while fewer models exhibit the

语言模型在宗教身份之间混淆亲属称谓。文化不合适的预测在模型之间显示出明显的方向性不对称。在所有实验中，模型被指示根据孟加拉国的文化习俗回答问题，使用的提示策略包括明确的宗教身份标记（例如，“一个穆斯林人”）、具有宗教暗示的个人姓名（例如，“Arnab Roy”，一个文化上属于印度教的姓名），以及没有明显宗教提示的开放性提示。一些模型，包括 GPT-4o、DeepSeek V3.1 和 Gemini 2.5 Flash，在明确或隐含的印度教语境中表现出更强的倾向，用与穆斯林相关的亲属称谓进行替代，而较少的模型表现出这种现象。

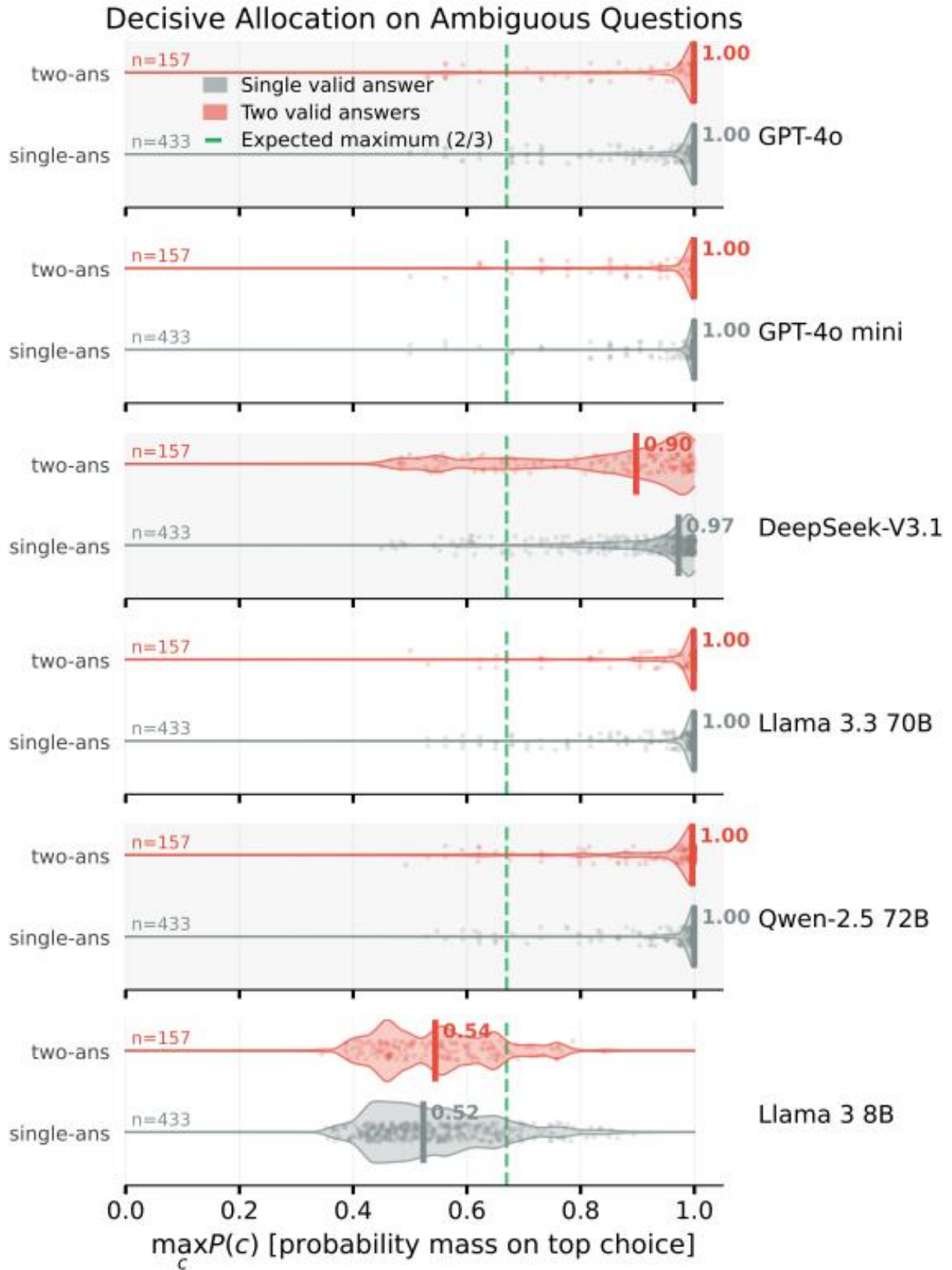


Figure 6: images/056619a34a6d15808930ce675bccd9ccb76be470b6be378edf336c596bac0a04.jpg

Figure 6: Probability mass allocation between ambiguous and unambiguous questions. Distribution of $\max_c P(c)$, the probability mass assigned to the most probable choice, over single-answer

and two-answer questions, recovered from token-level log-probabilities via choice-restricted softmax. A well-calibrated model should show lower probability mass on two-answer questions than on single-answer ones.

图 6: 模糊问题和明确问题之间的概率质量分配。 $\max_c P(c)$ 的分布, 即分配给最可能选项的概率质量, 在单选题和双选题上的分布, 通过限制选项的 softmax 从 token 级的对数概率中恢复得到。一个校准良好的模型应该在双选题上表现出比单选题更低的概率质量。

reverse pattern. To quantify this asymmetry, we perform a χ^2 goodness-of-fit test for each model over all culturally inappropriate predictions involving religious mismatch. The number of such cases varies by model ($N = 24-65$). Eleven of twelve models exhibit statistically significant asymmetry ($p < 0.05$), with a pronounced skew toward substituting Muslim-associated kinship terms in Hindu-marked contexts. Detailed results appear in Appendix D.10.1.

反向模式。为了量化这种不对称性, 我们对每个模型在所有涉及宗教不匹配的文化不适当预测中进行 χ^2 拟合优度检验。此类案例的数量因模型而异 ($N = 24-65$)。十二个模型中有十一个表现出统计学上显著的不对称性 ($p < 0.05$), 其中在以印度教为标志的语境中, 倾向于用与穆斯林相关的亲属称谓进行替换的现象尤为明显。详细结果见附录 D.10.1。

Log-probabilities reveal decisive probability allocation on ambiguous situations For models with accessible token-level log probabilities (Figure 6), we recover log-probabilities via choice-restricted softmax over pronoun logits using the top-k alternatives ($k = 5$) returned by the API (Appendix D.11.1), and compute $\max_c P(c)$ per question. Bangla tokenization inconsistencies were mitigated by mapping answer options to letter labels (A-C) and restricting model outputs to

对模糊情境的决定性概率分配可以通过对数概率来揭示对于可以访问到 token 级别的对数概率的模型 (图 6), 我们通过使用 API 返回的 top-k 替代选项 ($k = 5$) 对代词 logits 进行限制选择的 softmax, 从而恢复对数概率 (附录 D.11.1), 并按问题计算 $\max_c P(c)$ 。通过将答案选项映射到字母标签 (A-C) 并限制模型输出, 缓解了孟加拉语分词不一致的问题

these labels. Questions are split into unambiguous ($n=433$, single valid answer) and ambiguous ($n=157$, two valid answers). We use $\frac{2}{3} \approx 0.67$ as a reference boundary, above which one form holds more than twice the probability mass of any alternative. Figure 6 shows that five of six models produce median $\max_c P(c) = 1.00$ on ambiguous questions, identical to unambiguous ones, indicating they assign near-total probability mass to a single pronoun regardless of sociopragmatic ambiguity. Only Llama 3 8B deviates (median 0.54), though still near the 0.67 boundary. This suggests that current LLMs allocate probability mass as decisively on sociopragmatically ambiguous address forms as on unambiguous ones, without reflecting the variation acknowledged by human annotators.

这些标签。问题被分为明确的 ($n=433$, 单个有效答案) 和模糊的 ($n=157$, 两个有效答案)。我们使用 $\frac{2}{3} \approx 0.67$ 作为参考边界, 超过该边界的模型中, 一个形式的概率质量超过任何其他形式的两倍。图 6 显示, 六个模型中有五个在模糊问题上产生中位数 $\max_c P(c) = 1.00$, 与明确问题相同, 表明它们无论社会语用模糊性如何, 都将几乎全部的概率质量分配给一个代词。只有 Llama 3 8B 存在偏差 (中位数为 0.54), 尽管仍然接近 0.67 的边界。这表明当前的大型语言模型在社会语用模糊的地址形式上分配概率质量的方式与明确形式一样果断, 而没有反映人类标注员所认可的变化。

5 Conclusion

5 结论

Cultural competence in language models requires not only linguistic fluency but also sensitivity to sociopragmatic context. Using BANGLASOCIALBENCH, we show that even strong LLMs exhibit systematic limitations in Bangla address terms, kinship reasoning, and social customs, often defaulting to overly polite forms, lacking calibration in ambiguous contexts, and conflating kinship terminology across religious identities. Addressing the sociopragmatic weaknesses

exposed by our evaluation requires both data-centric and algorithmic interventions. We hypothesize that deficits such as systematic over-politeness and cross-religious kinship conflation stem from a training data imbalance. Current multilingual LLMs are primarily exposed to formal or translated Bangla corpora, leaving informal contexts, downward-hierarchy interactions, and minority-community norms systematically underrepresented. While training native Bengali LMs from scratch may offer deeper cultural embedding, the computational requirements for general reasoning parity remain prohibitive. A more pragmatic approach lies in culturally grounded alignment. Future research could utilize culturally specific adapters or Direct Preference Optimization (DPO) with contrastive pairs derived from the failure modes identified in BANGLASOCIALBENCH. Such methods would allow practitioners to leverage the reasoning capabilities of existing models while anchoring them more precisely to native Bengali relational norms.

语言模型的文化能力不仅需要语言流利，还需要对社会语用语境的敏感度。利用 BANGLASOCIALBENCH，我们表明，即使是强大的大语言模型（LLMs）在孟加拉语称呼用语、亲属关系推理和社会习俗方面也存在系统性的局限，常常默认使用过于礼貌的形式，在歧义语境中缺乏校准，并且在宗教身份之间混淆亲属术语。解决我们评估中暴露的社会语用弱点，需要数据驱动和算法层面的干预。我们假设系统性的过度礼貌和跨宗教亲属术语混淆等缺陷，源于训练数据的不平衡。目前的多语言大语言模型主要接触的是正式或翻译过的孟加拉语语料库，导致非正式语境、等级关系互动以及少数群体规范被系统性地低估。虽然从头训练本土孟加拉语语言模型可能提供更深层次的文化嵌入，但实现通用推理能力的计算需求仍然不可行。一种更具实用性的方法在于基于文化背景的对齐。未来的研究可以利用特定文化适配器或直接偏好优化（DPO），结合从 BANGLASOCIALBENCH 中识别出的失败模式中得出的对比对。这些方法将使从业者能够利用现有模型的推理能力，同时更精确地将其与本土孟加拉语关系规范对齐。

Limitations

限制

While BANGLASOCIALBENCH provides a focused evaluation of sociopragmatic competence in Bangladeshi social interaction, several limitations remain.

虽然 BANGLASOCIALBENCH 提供了对孟加拉国社交互动中社会语言能力的针对性评估，但仍存在一些局限性。

Standard Colloquial Bangla vs. dialect All instances and annotations are grounded in Standard Colloquial Bangla, reflecting widely shared norms of address usage and social interaction in Bangladesh. Annotators were instructed to judge cultural appropriateness with respect to this interactional standard. As a result, the benchmark does not capture the regional and dialectal variation present across Bangla varieties. Address pronouns, kinship terms, and honorifics may differ in regional varieties such as Sylheti, Chittagonian, or Rangpuri, and culturally appropriate usage in these dialects may not always align with the norms encoded in our dataset.

标准口语孟加拉语 vs. 方言所有实例和注释均基于标准口语孟加拉语，反映了孟加拉国广泛共享的称呼使用和社会互动规范。标注者被指示根据这一互动标准来判断文化上的恰当性。因此，基准数据并未捕捉到孟加拉语各种变体中存在的地域和方言差异。在如锡尔赫特语、奇塔冈尼亚语或朗普里语等地区变体中，称呼代词、亲属称谓和敬语可能会有所不同，这些方言中文化上恰当的使用方式并不总是与我们数据集中编码的规范一致。

We restrict the benchmark to Standard Colloquial Bangla for two reasons. First, it functions as a common interactional baseline used in interregional communication, education, media, and institutional settings. Second, regional dialects differ not only lexically but also in their pragmatic conventions, which would require dialect-specific annotation guidelines and expert annotators for each variety. Without sufficient dialectal coverage, such variation could be misinterpreted as model errors.

我们限制基准测试仅限于标准孟加拉语，有两个原因。首先，它作为跨地区交流、教育、媒体和机构设置中使用的共同互动基准。其次，区域方言不仅在词汇上有所不同，而且在语用惯例上也存在差异，这需要

为每种方言制定特定的注释指南和专家注释者。如果没有足够的方言覆盖，这种变化可能会被误认为是模型错误。

Coverage of interactional contexts Although the dataset spans multiple sociopragmatic settings, including family, office, hospital, academic, and everyday social interactions, it cannot exhaustively represent all interactional configurations present in Bangladeshi society. Factors such as rural-urban differences, sociolectal variation, and code-switching practices are not explicitly modeled.

互动语境的覆盖范围尽管数据集涵盖了多种社会语用情境，包括家庭、办公室、医院、学术以及日常社会互动，但它无法全面呈现孟加拉国社会中所有互动配置。诸如城乡差异、社会方言变化以及语言转换实践等因素并未明确建模。

Consequently, models that generate regionally appropriate but non-standard forms may be penalized under our evaluation framework. This should not be interpreted as a claim that Standard Colloquial Bangla is inherently more correct, but rather as a methodological constraint required to ensure annotation consistency and comparability across models. Future work may extend the benchmark to incorporate dialectal and regional sociopragmatic variation.

因此，在我们的评估框架下，生成符合地区特点但非标准形式的模型可能会受到惩罚。这不应被理解为声称标准口语孟加拉语本质上更加正确，而应被视为一种方法上的限制，以确保模型之间注释的一致性和可比性。未来的工作可能会扩展基准测试，以纳入方言和地域社会语言学变异。

Ethical Considerations

伦理考量

Annotators were compensated for their work at a rate consistent with local standards for similar annotation tasks. Prior to participation, annotators were informed about the purpose of the study and how their contributions would be used in the dataset.

标注人员的报酬与其所在地区类似标注任务的常规标准一致。在参与之前，标注人员被告知了研究的目的以及他们的贡献将如何用于数据集中。

To protect privacy, annotators were instructed not to include any personally identifiable information when writing scenarios. All collected instances were additionally reviewed by the research team to ensure that they did not contain harmful, unsafe, or sensitive content. The benchmark focuses on culturally grounded everyday interactions and does not include sexual, violent, or personally identifiable material.

为保护隐私，标注人员在编写场景时被指示不要包含任何个人可识别信息。所有收集到的实例还由研究团队进行了额外的审查，以确保其中不包含有害、不安全或敏感的内容。该基准测试聚焦于文化背景下的日常互动，不包括色情、暴力或包含个人可识别信息的内容。

References

参考资料

Muhammad Farid Adilazuarda, Sagnik Mukherjee, Pradhyumna Lavania, Siddhant Singh, Alham Fikri Aji, Jacki O'Neill, Ashutosh Modi, and Monojit Choudhury. 2024. Towards Measuring and Modeling “Culture” in LLMs: A Survey. Preprint, arXiv:2403.15412.

Badr AlKhamissi, Muhammad ElNokrashy, Mai Alkhamissi, and Mona Diab. 2024. Investigating Cultural Alignment of Large Language Models. In Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 12404–12422, Bangkok, Thailand. Association for Computational Linguistics.

- Ron Artstein and Massimo Poesio. 2008. Survey Article: Inter-Coder Agreement for Computational Linguistics. *Computational Linguistics*, 34(4):555–596.
- Lama Ayash, Hassan Alhuzali, Ashwag Alasmari, and Sultan Aloufi. 2025. SaudiCulture: A Benchmark for Evaluating Large Language Models Cultural Competence within Saudi Arabia. Preprint, arXiv:2503.17485.
- Anne Barron. 2003. Acquisition in Interlanguage Pragmatics: Learning How to Do Things with Words in a Study Abroad Context. Ph.D. thesis, J. Benjamins, Amsterdam Philadelphia (Pa.).
- Friederike Braun. 1988. Terms of Address: Problems of Patterns and Usage in Various Languages and Cultures. DE GRUYTER.
- Roger Brown and Albert Gilman. 1968. THE PRONOUNS OF POWER AND SOLIDARITY. In Joshua A. Fishman, editor, *Readings in the Sociology of Language*, pages 252–275. DE GRUYTER.
- Xinren Chen and Juanjuan Ren. 2020. A memetic cultural practice: The use of generalized kinship terms in a research seminar attended by Chinese graduate students. *Lingua*, 245:102942.
- Jacob Cohen. 1960. A Coefficient of Agreement for Nominal Scales. *Educational and Psychological Measurement*, 20(1):37–46.
- Sisir Kumar Das. 1968a. Forms of address and terms of reference in Bengali. *Anthropological Linguistics*, 10(4):19–31.
- Sisir Kumar Das. 1968b. Forms of Address and Terms of Reference in Bengali. *Anthropological Linguistics*, 10(4):19–31.
- Eleanor Dickey. 1997. Forms of address and terms of reference. *Journal of Linguistics*, 33(2):255–274.
- Mohammad Rifqi Farhansyah, Iwan Darmawan, Adryan Kusumawardhana, Genta Indra Winata, Alham Fikri Aji, and Derry Tanti Wijaya. 2025. Do Language Models Understand Honorific Systems in Javanese? In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 26732–26754, Vienna, Austria. Association for Computational Linguistics.
- Cliff Goddard and Anna Wierzbicka. 2004. Cultural scripts: What are they and what are they good for? *Intercultural Pragmatics*, 1(2).
- Dell Hymes. 1964. Introduction: Toward Ethnographies of Communication¹. *American Anthropologist*, 66(6_PART2):1–34.
- D.H. Hymes. 1962. The Ethnography of Speaking. *The Ethnography of Speaking*. Anthropological Society of Washington.
- Saman Sarker Joy. 2025. BnMMLU: Measuring Massive Multitask Language Understanding in Bengali. Preprint, arXiv:2505.18951.
- Daeen Kabir, Minhajur Rahman Chowdhury Mahim, Sheikh Shafayat, Adnan Sadik, Arian Ahmed, Eunsu Kim, and Alice Oh. 2025. BLUCK: A Benchmark Dataset for Bengali Linguistic Understanding and Cultural Knowledge. Preprint, arXiv:2505.21092.
- Mohammad Hossein Keshavarz. 2001. The role of social context, intimacy, and distance in the choice of forms of address. *International Journal of the Sociology of Language*, 2001(148).
- Eunsu Kim, Juyoung Suk, Philhoon Oh, Haneul Yoo, James Thorne, and Alice Oh. 2024. CLICK: A Benchmark Dataset of Cultural and Linguistic Intelligence in Korean. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 3335–3346, Torino, Italia. ELRA and ICCL.
- Muhammad Farid Adilazuarda, Sagnik Mukherjee, Pradhyumna Lavania, Siddhant Singh, Alham Fikri Aji, Jacki O’Neill, Ashutosh Modi, and Monojit Choudhury. 2024. 面向测量和建模“文化”在 LLMs 中的调查。预印本, arXiv:2403.15412.
- Badr AlKhamissi, Muhammad ElNokrashy, Mai Alkhamissi, and Mona Diab. 2024. 探究大型语言模型的文化一致性。在第 62 届计算语言学协会年会论文集 (第 1 卷: 长论文), 第 12404–12422 页, 泰国曼谷。计算语言学协会。
- Ron Artstein and Massimo Poesio. 2008. 调查文章: 计算语言学中的编码者间一致性。计算语言学, 34(4):555–596。
- Lama Ayash, Hassan Alhuzali, Ashwag Alasmari, and Sultan Aloufi. 2025. SaudiCulture: 评估大型语言模型在沙特阿拉伯文化能力的基准。预印本, arXiv:2503.17485。
- Anne Barron. 2003. 习得在中介语言语用学: 学习如何在海外学习环境中使用词语做事。博士学位论文, J. Benjamins, 阿姆斯特丹费城 (宾夕法尼亚州)。
- Friederike Braun. 1988. 称谓术语: 各种语言和文化中模式和使用的的问题。DE GRUYTER。
- Roger Brown and Albert Gilman. 1968. 权力与团结的代词。在 Joshua A. Fishman 编辑的《语言社

会学阅读》，第 252–275 页。DE GRUYTER。

Xinren Chen and Juanjuan Ren. 2020. 一种文化实践：中国研究生参加的研究研讨会中使用普遍亲属称谓。Lingua, 245:102942。

Jacob Cohen. 1960. 命名尺度的一致性系数。教育与心理测量, 20(1):37–46。

Sisir Kumar Das. 1968a. 孟加拉语中的称谓形式和参考术语。人类学语言学, 10(4):19–31。

Sisir Kumar Das. 1968b. 孟加拉语中的称谓形式和参考术语。人类学语言学, 10(4):19–31。

Eleanor Dickey. 1997. 称谓形式和参考术语。语言学杂志, 33(2):255–274。

Mohammad Rifqi Farhansyah, Iwan Darmawan, Adryan Kusumawardhana, Genta Indra Winata, Alham Fikri Aji, and Derry Tanti Wijaya. 2025. 语言模型是否理解爪哇语的敬语系统？在第 63 届计算语言学协会年会论文集（第 1 卷：长论文），第 26732–26754 页，奥地利维也纳。计算语言学协会。

Cliff Goddard and Anna Wierzbicka. 2004. 文化脚本：它们是什么，有什么用？跨文化语用学, 1(2)。

Dell Hymes. 1964. 引言：迈向交流的民族志¹。美国人类学家, 66(6_PART2):1–34。

D.H. Hymes. 1962. 说话的民族志。说话的民族志。华盛顿人类学学会。

Saman Sarker Joy. 2025. BnMMLU：测量孟加拉语中的大规模多任务语言理解。预印本, arXiv:2505.18951。

Daeen Kabir, Minhajur Rahman Chowdhury Mahim, Sheikh Shafayat, Adnan Sadik, Arian Ahmed, Eunsu Kim, and Alice Oh. 2025. BLUCK：孟加拉语语言理解和文化知识的基准数据集。预印本, arXiv:2505.21092。

Mohammad Hossein Keshavarz. 2001. 社会背景、亲密关系和距离在选择称谓形式中的作用。语言社会学国际期刊, 2001(148)。

Eunsu Kim, Juyoung Suk, Philhoon Oh, Haneul Yoo, James Thorne, and Alice Oh. 2024. CLICK：韩国文化与语言智能的基准数据集。在 2024 年联合国际计算语言学、语言资源和评估会议（LREC-COLING 2024）论文集，第 3335–3346 页，意大利都灵。ELRA 和 ICCL。